

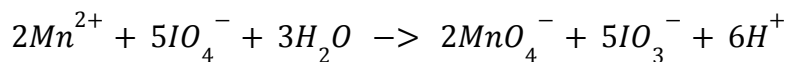
Beer's Law Determination of Mn in Steel via Spectrophotometry

Abstract:

A sample of steel was analyzed to determine the mass percentage of the analyte, manganese (Mn^{2+}), using spectrophotometry. Two solutions were prepared: a standard steel sample Mn^{2+} was diluted into different concentrations, and their absorbances were measured at λ_{max} (525nm) of the permanganate ion. The unknown solution contained a steel sample with an unknown concentration of Mn. The absorbances were plotted using Beer's law $A = \epsilon bC$, establishing a linear plot. The equation of this graph was taken, and the concentration of the sample was determined using the absorbance of the unknown solution. Through reverse computations, the value of 0.288% Mn^{2+} was determined, which closely matches the accepted value of 0.28%.

Introduction:

Manganese undergoes a distinct color change when oxidized from its initial 2+ Manganese colorless oxidation state, to oxidizing it to the permanganate ion (MnO_4^-). The permanganate ion has a dark purple color, which can be used to measure the absorbance via spectrophotometry. The oxidation is achieved through the use of potassium periodate (KIO_4) in strongly acidic conditions, described by the reaction:



The acidic environment (using phosphoric and nitric acid) minimizes the risk of forming colloidal MnO_2 – which is observed under less controlled conditions (1). This ensures that the

Mn^{2+} solution was fully converted to the colored permanganate ion, enabling more precise absorbance measurements. Alternative oxidation strategies have also been explored, such as on-line electrochemical oxidation using Ag^+ catalysis, which has been shown to further enhance the oxidation of Mn^{2+} to MnO_4^- (2). The developed color of the permanganate ion allows for the use of UV/Vis spectrophotometry. Beer's law ($A = \epsilon bC$) (in which A is absorbance, C is the concentration of the absorbing species, ϵ is molar absorptivity, and b is the optical path length (3)) allows for the creation of a linear graph, in the form $y = mx + b$ in which y is equal to the absorbance of the analyte, and x is equal to the concentration, or the molarity of Mn in the sample. The pros of spectrophotometry are that it requires a very small amount of sample and is a well-validated, replicable analytical technique. The cons of this method are that it can only operate on samples within the 200-800 nm range, and it is highly sensitive to external noise. It also requires a similar standard solution to be established before the analysis of the sample. This experiment aimed to determine the mass % of Mn in an unknown steel sample by comparing the absorptivity with a known standard at various concentrations.

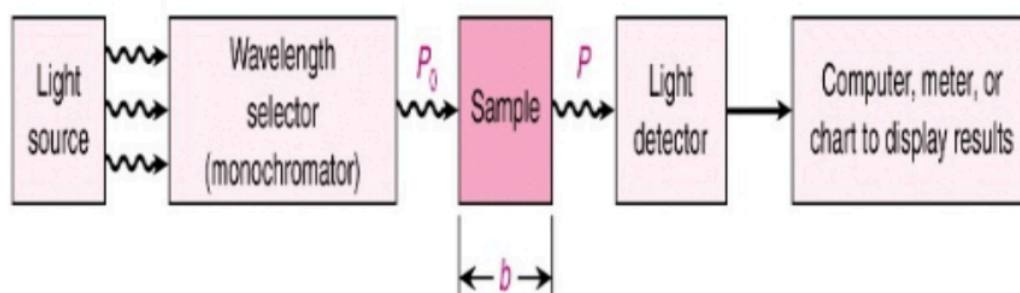


Figure 1: Block Diagram of a Spectrophotometer

The samples used in this experiment were measured using a spectrophotometer. Each sample was placed into a cuvette, then put in the middle slot shown in Figure 1 labeled sample. UV light is then beamed from the light source through the sample. The light detector then measures the intensity of the light passing through the sample and compares it to the initial intensity to determine the absorbance of the sample. The absorbance of the sample can be used to calculate the concentration of the analyte.

Experimental:

A steel sample (0.5-0.6g) with a known manganese concentration was weighed and dissolved in 1:3 nitric acid. The solution was boiled, and ammonium persulfate was added. It was diluted to 100 mL using 18MΩ-cm water and boiled again. 85% phosphoric acid and potassium periodate were added to oxidize Mn^{2+} to MnO_4^- , then it was diluted to 200 mL and separated into different concentrations. The unknown manganese concentration steel sample was prepared similarly. A spectrophotometer measured the absorbances, which were used to create a Beer's law plot. The unknown sample's absorbance was then compared to the plot to determine the Mn concentration.

Results:

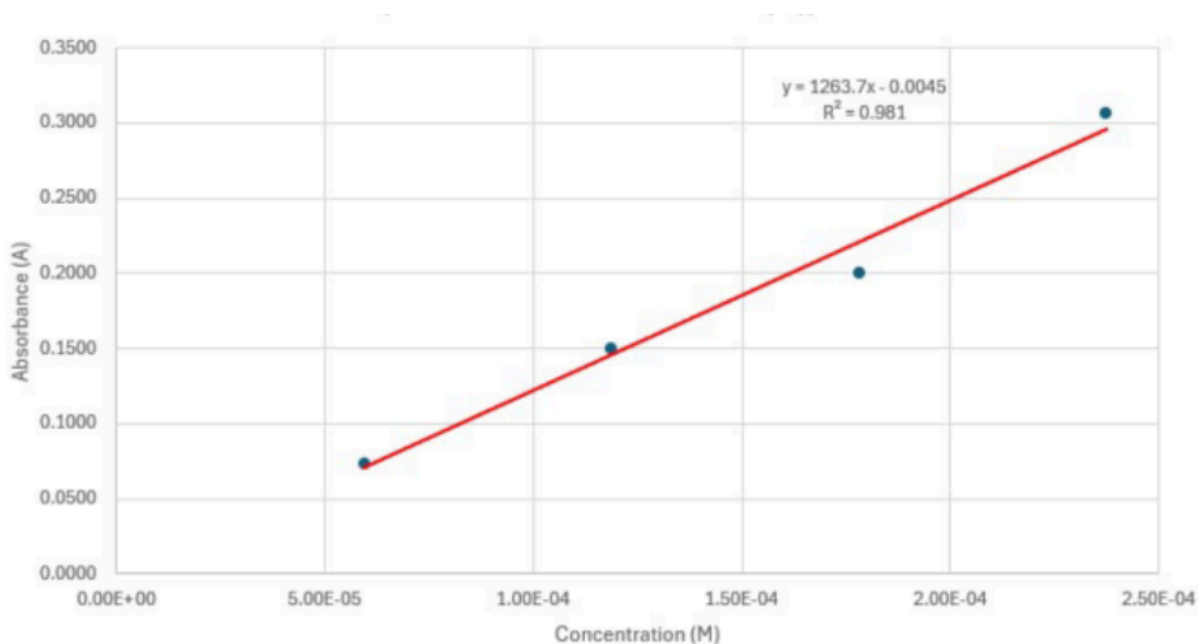
A linear calibration curve was generated from the results of the four standard dilutions. After applying the equation generated from the plot to the absorbance of the unknown sample, a value of 28.7% Mn was yielded.

Sample ID	True Value (%Mn)	Initial Volume of Steel (g)
#68 (standard)	0.511	0.6383
#75 (unknown)	0.280	0.6225

Table 1: Given values and initial measurements of steel samples.

Analyte	Absorbance (A) output at $\lambda 525$
Dilution of standard 1(100%)	0.307
Dilution of standard 2(75%)	0.230
Dilution of standard 3(50%)	0.150
Dilution of standard 4(25%)	0.074
Unknown Sample	0.201

Table 2: Data recorded from the spectrophotometer of standard dilutions and unknowns.



Graph #1: Spectrophotometric Determination of Mn in Standard of Steel (Absorbance vs Concentration(M)), Beer's Law plot for the absorbance of the diluted standard solutions.

To establish the Beer's law plot, the standard was diluted into four concentrations. The plot $y = 1263.7x - 0.0045$ was established. Using the absorbance value of 0.201, the concentration (x) was calculated to be 1.64×10^{-4} , which is the Moles of Mn. The moles of Mn were determined by multiplying the molarity by the volume

$1.63 \times 10^{-4} M \times 0.200 L = 3.25 \times 10^{-5} \text{ mols}$. Then, multiplied by the m/w of Manganese

$3.25 \times 10^{-5} \text{ mol} \times 54.94 \text{ g/mol} = 1.79 \times 10^{-3}$ grams of Manganese in the unknown

sample. This can be divided into the total starting sample to determine the percentage of Mn in

the sample $\frac{1.79 \times 10^{-3} \text{ g}}{0.6225} \times 100\% = 0.287\%$.

Discussion and Conclusion:

From the experiment, the results indicated that the manganese content in the steel sample was approximately 28.7%, which compares closely with the true value of the sample, 28.0%. To determine this value, a calibration curve was constructed using the absorbance values of the standard solutions measured at 525 nm. A linear regression of this graph resulted in the equation in the form $y = mx + b$, where the slope(m) allows for the relation of the measured absorbance (y) to the concentration (x) of Mn. Using the measured absorbance from the unknown sample, the molarity of Mn was calculated, and after converting to grams using the molecular weight of manganese, the mass percentage was determined by dividing by the initial sample mass.

This experiment yielded a relative error of 25 ppt, which is a strong suggestion of random error rather than a systematic error in the experiment. Since this deviation is small, the sources of error are likely from slight measurement fluctuations, a minor variation in the sample, or differences in steps such as boiling the solution. While systematic errors (such as instrumental

calibration or procedural loss) were not apparent, if present, they would also affect the final result proportionally.

The spectrophotometric method used in this experiment offers several advantages: it requires a small sample size and has been widely validated as an analytical technique.

Disadvantages include the sensitivity to external noise, temperature fluctuations of the sample, and its limited range of 200-800nm (4). These are factors that introduce variability, limiting the accuracy of the procedure. Alternative quantitative methods for manganese analysis, such as atomic absorption spectroscopy, offer improved sensitivity and massively reduce interference (5). This increase in sensitivity is reflected in the more complex instrumentation and operational costs.

Overall, the proximity of the experimental and accepted values validates the method of spectrophotometry. Further improvements could include a more rigorous calibration, and a consistent temperature between the two samples would help minimize noise to minimize random and systematic error.

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Peer Review Evaluation:

The peer review for this assignment provided valuable insight into what changes needed to be made in my initial draft. Many of the mistakes made in the initial draft were caught in this proofread, including significant figure mistakes and the use of different tenses. Edits on the graph were made, removing titles from the draft and including them in the caption. The peer review also allowed me to reflect on the things that I did well, such as the abstract, as I believe my initial draft contained a decent template for the final paper. The inclusion of more sources allowed for the validation of many claims presented in the paper. Overall, the peer review provided valuable insight that strongly influenced my final draft.